VIKUM BANDARA

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Mathematics researcher with an M.S. in Mathematical Sciences and experience using Python to investigate discrete systems, test conjectures, and develop algorithms. Foundation in probability, linear algebra, and optimization, supported by coauthored research and university teaching. Seeking to apply this approach to quantitative research in financial markets.

Education

Clemson University Clemson, SC, USA
M.S. in Mathematical Sciences December 2025
Continuing doctoral study in Mathematical Sciences; cumulative graduate GPA: 3.57/4.00.
GPA includes coursework through Spring 2026./ Passed PhD candidacy preliminary exam in Operations Research.
Selected coursework: Probability, Stochastic Processes, Data Analysis, Matrix Analysis, Mathematical Programming, Advanced Linear Programming, Discrete Optimization, and Network Flow Programming.

University of Peradeniya Peradeniya, Sri Lanka
B.Sc. (Hons) in Mathematics, September 2022
Overall GPA: 3.66/4.00; Mathematics GPA: 3.77/4.00. Minor in Physics.
Selected coursework: Linear Algebra, Real Analysis, Probability Theory, and Financial Mathematics.

Skills and Trading Experience

Python: computational experiments, visualization, conjecture testing, and encryption algorithms.
Additional tools (familiar): MATLAB, Julia, Gurobi, RStudio. **Technical writing:** LaTeX.
Personal trading: Experience trading cryptocurrencies and options on Binance.
Languages: English and Sinhala; university teaching and research communication in English.

Computational Research

Combinatorics and Discrete Dynamics Clemson University
Research with Dr. Svetlana Poznanović on Bulgarian solitaire with distinguishable cells.

- Used Python to visualize orbits and identify when configuration shapes repeat before individual labeled cells return to their original positions.
- Developed a coloring algorithm on Ferrers diagrams connecting labeled-cell movement to permutation cycles and orbit lengths.
- Coauthored a preprint establishing periodicity criteria and exact orbit sizes for four labeled variants of Bulgarian solitaire.

Cryptography and Graph Theory University of Peradeniya, 2021–2023

- Implemented proposed Collatz-based encryption and decryption algorithms in Python; coauthored a journal publication in 2026.
- Investigated graph-labeling constructions for symmetric encryption using snake graphs and supermagic coverings; coauthored a journal publication in 2024.

Selected Publications and Preprints

- **V. Bandara** and S. Poznanović. "Bulgarian Solitaire with distinguishable cells." *Preprint* (Master's project manuscript).
- "Collatz conjecture–driven cryptographic framework." *Journal of Discrete Mathematical Sciences & Cryptography*, 2026.
- "Symmetric encryption using snake graphs and supermagic covering." *Journal of the National Science Foundation of Sri Lanka*, 2024.

Teaching and Communication

Graduate Teacher of Record, Clemson University January 2025–Present
Teach Business Calculus I (MATH 1020), explaining mathematical concepts in a business-focused course.

Graduate Teaching Assistant, Clemson University August 2023–December 2024
Supported courses in precalculus, calculus, ordinary differential equations, and linear algebra.

Teaching Assistant, University of Peradeniya September 2022–April 2023

Presentations and coordination: RESCON Research Conference and SURS, 2022.